STATISTICAL ANALYSIS REPORT

Data Preparation and Statistical Analysis of Student Performance Using SPSS

— Applied Statistics and Data Analysis —

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Course: Applied Statistics and Data Analysis (MTC)

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1. Executive Summary

Report Overview

This report presents a complete statistical analysis of a student performance dataset (n = 20) using SPSS. It covers data preparation, outlier detection, descriptive statistics, inferential testing, and correlation analysis — with the aim of understanding the relationships between gender, study hours, GPA, and attendance.

Objectives

- Prepare the dataset: recode string variables, handle missing data, detect and treat outliers
- Compute descriptive statistics for Study_Hours, GPA, and Attendance, disaggregated by gender
- Test whether GPA differs significantly between male and female students (independent-samples t-test)
- Quantify the linear association between weekly study hours and GPA (Pearson correlation)
- Present findings through professional tables and embedded SPSS output charts

Key Findings at a Glance

Finding	Result	Statistical Status
Missing GPA values	1 of 20 (5%)	Excluded via listwise deletion
Outlier — Study Hours	Case #19: 200 hrs/week	Filtered; sensitivity analysis run
Female mean GPA	3.41	Slightly higher than males
Male mean GPA	3.12	Not significantly different
GPA gender difference	$t(16) = -1.44$, $p = .170$	Not statistically significant
Study Hours vs. GPA	$r = .876$, $p < .001$	Strong positive correlation ★

Overall: study effort is a strong predictor of GPA, while gender differences are not statistically significant in this sample. Behavioral factors (study time) outweigh demographic factors in determining academic performance.

2. Data Description and Preparation

2.1 Dataset Overview

The dataset comprises 20 student records containing five variables:

Variable	Type	Description	Measurement Scale
ID	Numeric	Unique student identifier	Nominal
Gender	String → recoded	Student gender (Male / Female)	Nominal
Study_Hours	Numeric	Weekly hours devoted to studying	Scale (Ratio)
GPA	Numeric	Grade Point Average (0.0 – 4.0)	Scale (Ratio)
Attendance	Numeric	Class attendance percentage (0 – 100)	Scale (Ratio)

2.2 Handling String Variables — Gender Recoding

The Gender variable was stored as text and recoded to numeric using the following SPSS syntax:

```
RECODE Gender ('Male'=1) ('Female'=2) INTO Gender_num.
VARIABLE LABELS Gender_num '1 = Male, 2 = Female'.
EXECUTE.
```

Recoding Note

Male = 1, Female = 2. The original string variable was retained. This numeric coding is required by parametric tests (t-test, ANOVA) which need numeric group identifiers.

2.3 Missing Values Analysis (GPA)

A FREQUENCIES analysis was run on GPA to identify missing data before any inferential tests.

Category	N	Percent
Valid GPA observations	19	95.0%
System-missing (no GPA recorded)	1	5.0%
Total cases in dataset	20	100.0%

Missing Data Decision

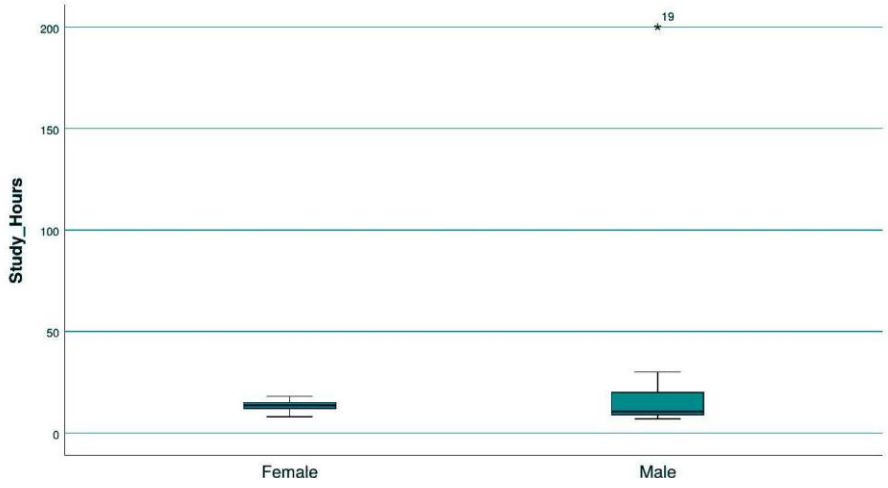
GPA has 19 valid observations and 1 system-missing value (5% of the sample). The missing case was excluded only from analyses involving GPA, and all other variables for this student were kept. Listwise deletion was applied for GPA-specific analyses, preserving maximum data for non-GPA procedures.

3. Outlier Detection (Study Hours)

3.1 First Boxplot — Full Dataset (N = 20)

A side-by-side boxplot of Study_Hours by Gender was produced. The extreme outlier is immediately visible as a star (*) far above the male distribution.

Figure 1 | Boxplot of Study_Hours by Gender — Unfiltered (N = 20)



Note: *Case #19 (Male) marked with * at 200 hours/week — an extreme outlier far beyond all other values (7–30 hrs).*

3.2 Detailed EXAMINE with Descriptives

A second EXAMINE run with stem-and-leaf plots and full descriptive statistics confirmed the outlier and its distorting effect on the male group.

Figure 2 | Boxplot of Study_Hours by Gender — EXAMINE with Descriptives

```
1.00        3 .  0
1.00 Extremes    (>=200)

Stem width:   10
Each leaf:        1 case(s)
```

Note: *Same outlier confirmed. Stem-and-leaf for males lists extreme as '≥200'. All other male values: 7, 8, 9, 10, 10, 11, 12, 20, 30.*

Outlier Finding

Stem-and-leaf plots and boxplots reveal an extreme outlier in Study_Hours: 200 hours per week for a male student (Case #19). All remaining values lie between 7 and 30 hours. Because 200 hrs/week (≈29 hrs/day) is impossible, it represents a data entry error. Additional analyses were repeated with a filter (Study_Hours < 50) to check robustness of results.

3.3 Impact on Descriptive Statistics

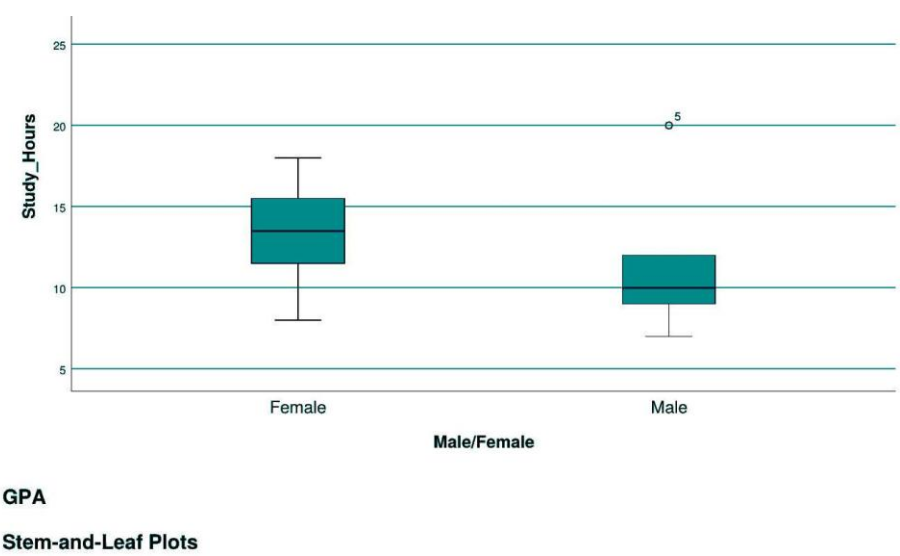
Statistic	Female (n=10)	Male — Full (n=10)	Male — Filtered (n=9)
Mean	13.40 hrs	31.70 hrs	13.00 hrs
Median	13.50 hrs	10.50 hrs	10.00 hrs
Std. Deviation	2.757 hrs	59.545 hrs	7.399 hrs
Minimum	8 hrs	7 hrs	7 hrs
Maximum	18 hrs	200 hrs	30 hrs
Skewness	-0.363	3.084	1.899

The outlier inflates the male mean from 13.0 to 31.7 hours and the standard deviation from 7.4 to 59.5 — a 706% inflation.

3.4 Sensitivity Filter

```
COMPUTE filter_$=(Study_Hours < 50).
VARIABLE LABELS filter_$ 'Study_Hours < 50 (FILTER)'.
FILTER BY filter_$.
EXECUTE.
```

Figure 3 | Boxplot of Study_Hours by Gender — Filtered Dataset (N = 19)



Note: *After filtering, both groups are comparable (≈10–13 hrs/week). Two mild outliers at cases 5 (20 hrs) and 9 (30 hrs) remain in the male group.*

4. Descriptive Statistics

4.1 Overall Statistics (Filtered, N = 19)

Variable	N	Minimum	Maximum	Mean	Std. Deviation
Study_Hours	19	7	30	13.21	5.308
GPA	18	2.5	4.0	3.267	0.4393
Attendance	18	65	98	83.94	9.334

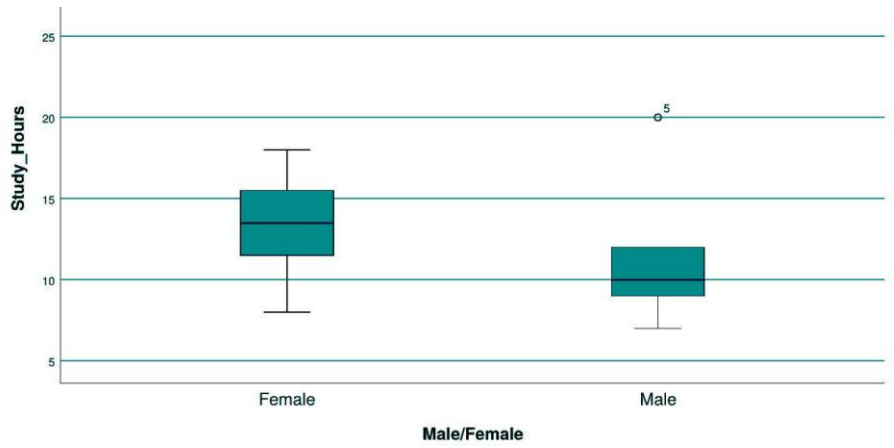
The sample shows strong academic indicators: mean GPA 3.27/4.0, mean attendance 83.9%, and an average study load of ~13 hrs/week.

4.2 Statistics by Gender

Study Hours

Statistic	Female (n=8)	Male (n=9)
Mean	13.38 hrs	13.00 hrs
Median	13.50 hrs	10.00 hrs
Std. Deviation	3.114 hrs	7.399 hrs
Range	8 – 18 hrs	7 – 30 hrs
Skewness	-0.314	1.899

Figure 4 | Boxplot of Study_Hours by Gender — Descriptive Comparison (Filtered)



GPA

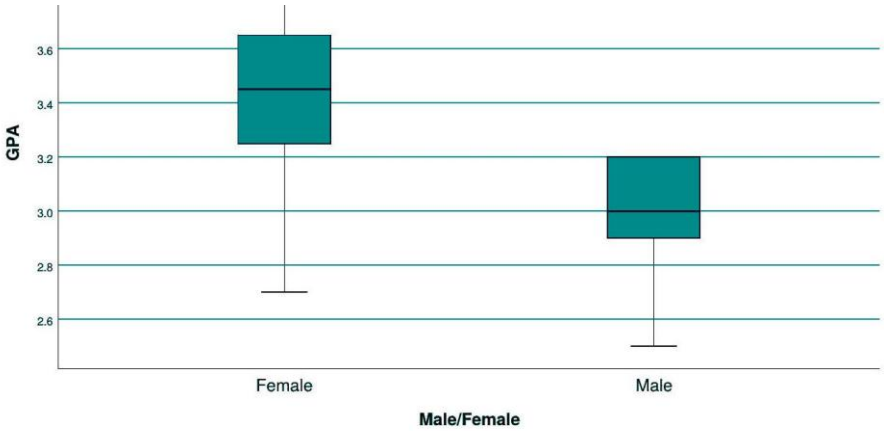
Stem-and-Leaf Plots

Note: *Female distribution is compact and symmetric (SD = 3.1); male distribution is positively skewed with larger spread (SD = 7.4).*

GPA by Gender

Statistic	Female (n=8)	Male (n=9)
Mean	3.412	3.122
Median	3.450	3.000
Std. Deviation	0.3643	0.4969
95% CI for Mean	[3.108, 3.717]	[2.740, 3.504]
Skewness	-0.890	0.773

Figure 5 | Boxplot of GPA by Gender



Attendance

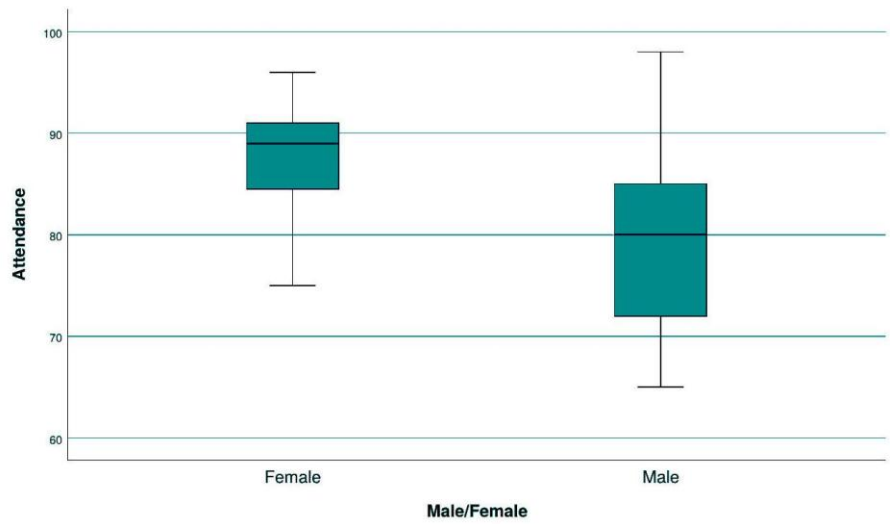
Stem-and-Leaf Plots

Note: *Female median GPA (3.45) > Male median GPA (3.00). Two mild upper outliers visible in male group (cases 5 and 9).*

Attendance by Gender

Statistic	Female (n=8)	Male (n=9)
Mean	87.50%	80.33%
Median	89.00%	80.00%
Std. Deviation	6.325%	10.989%
Range	75 – 96%	65 – 98%

Figure 6 | Boxplot of Attendance by Gender



Note: *Females show higher and more consistent attendance (SD = 6.3% vs 11.0%). Male attendance is more variable.*

Descriptive Insight

Across all outcomes, females perform slightly better: higher GPA (+0.29 points), higher attendance (+7.2%), and more consistent study hours. These differences are formally tested in Section 5.

5. Independent-Samples t-Test: GPA by Gender

5.1 Hypotheses

- H_0: No significant difference in mean GPA between males and females (μ_male = μ_female)
- H_1: Significant difference exists (μ_male ≠ μ_female); α = .05, two-tailed

5.2 Group Statistics

Group	N	Mean GPA	Std. Deviation	Std. Error Mean
Male	9	3.122	0.4969	0.1656
Female	9	3.411	0.3408	0.1136

5.3 Levene's Test + t-Test Results

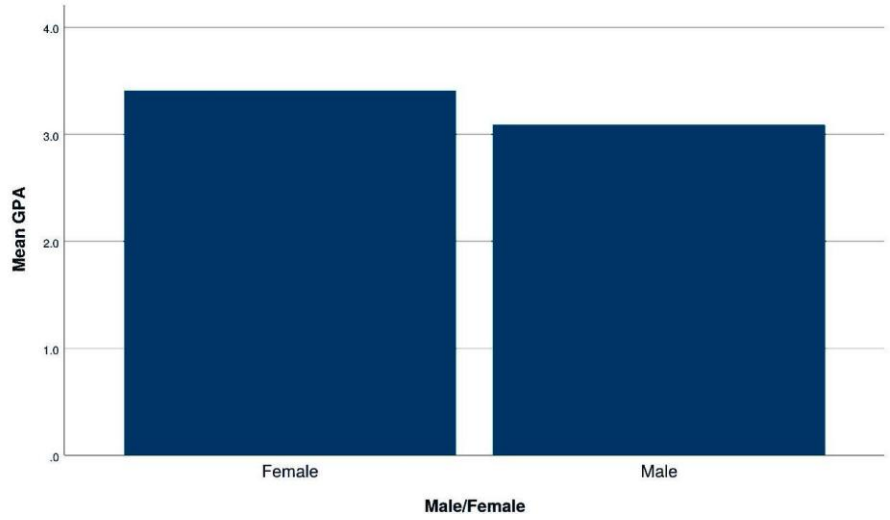
	Levene F	Levene p	t	df	p (2-tailed)	Mean Diff.	95% CI
Equal var. assumed	0.970	.339	-1.438	16	.170	-0.2889	[-0.715, 0.137]
Equal var. not assumed	—	—	-1.438	14.16	.172	-0.2889	[-0.719, 0.141]

Levene's test: F = 0.970, p = .339 (non-significant) → equal variances assumed. Use top row.

5.4 Effect Sizes

Measure	Point Estimate	95% CI Lower	95% CI Upper
Cohen's d	-0.678	-1.621	0.285
Hedges' correction	-0.646	-1.544	0.271
Glass's delta	-0.848	-1.836	0.184

Figure 7 | Bar Chart: Mean GPA by Gender



Note: *Female mean GPA (3.41) is visually higher than male (3.12), but the difference is not statistically significant (p = .170).*

t-Test Interpretation

t(16) = −1.44, p = .170. The GPA difference (males M = 3.12, SD = 0.50; females M = 3.41, SD = 0.34) is not significant at α = .05. We fail to reject H_0. Cohen's d = −0.68 indicates a medium effect, but the small per-group n = 9 provides insufficient power to detect an effect this size. A larger sample is needed.

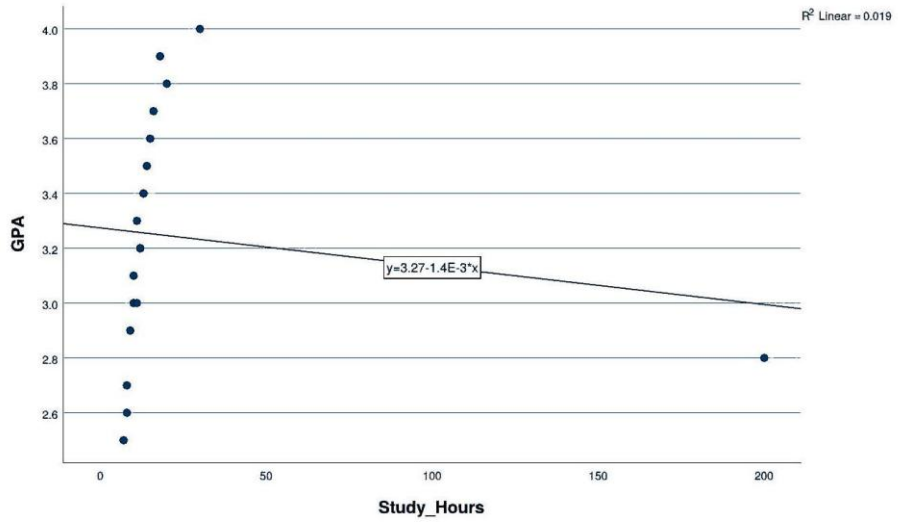
6. Correlation: Study Hours and GPA

6.1 Pearson Correlation Matrix

	Study_Hours	GPA
Study_Hours (r)	1.000	.876**
GPA (r)	.876**	1.000
N	19 / 18	18 / 18
Sig. (2-tailed)	—	.000

*** Correlation significant at 0.01 level (2-tailed).*

Figure 8 | Scatterplot: Study_Hours vs. GPA with Linear Fit (Full Dataset)



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Note: *Linear fit ($y = 3.27 - 1.4\times10^{-3}x$, $R^2 = 0.019$) appears flat due to the extreme 200-hr outlier. After filtering, $r = .876$ reveals the true strong positive relationship.*

Correlation Interpretation

Study_Hours and GPA are strongly and positively correlated: $r = .876$, $n = 18$, $p < .001$ (filtered dataset). $r^2 \approx .77$ — approximately 77% of GPA variance is explained by study hours. This is the strongest finding in the analysis. The full-dataset scatterplot shows $R^2 = 0.019$, demonstrating how severely one outlier suppressed the true correlation.

7. Overall Conclusion

The dataset of 20 students was thoroughly prepared and analysed in SPSS. Key operations:

- Recoded the string Gender variable to numeric format for parametric procedures
- Identified one system-missing GPA value (5%) and applied listwise deletion
- Detected an extreme outlier in Study_Hours (200 hrs/week, Case #19); filtered for sensitivity analysis
- Descriptive statistics: females slightly outperform males on GPA and attendance
- Independent-samples t-test: GPA gender difference not statistically significant ($p = .170$)
- Pearson correlation: strong, significant positive Study_Hours–GPA relationship ($r = .876$, $p < .001$)

Key Takeaway

Study effort is the dominant predictor of academic performance. Gender is not a significant GPA predictor, suggesting behavioral factors outweigh demographic factors. Future longitudinal research with larger samples should confirm these findings and examine potential gender-moderating effects.

7.1 Limitations

- Small sample size (N = 20) limits statistical power and generalizability
- One extreme outlier required exclusion, reducing the male group to n = 9
- Cross-sectional design prevents causal inference
- Correlation of r = .876 may partly reflect confounds (prior ability, course difficulty)

References & Software

IBM Corp. (2023). IBM SPSS Statistics for Windows, Version 29.0. Armonk, NY: IBM Corp.

Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Lawrence Erlbaum.

Field, A. (2018). Discovering Statistics Using IBM SPSS Statistics (5th ed.). SAGE Publications.

Tabachnick, B. G., & Fidell, L. S. (2019). Using Multivariate Statistics (7th ed.). Pearson.

End of Report

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